# Chapter 6: Project Schedule Management

**Information Technology Project Management, Ninth Edition**

Note: See the text itself for full citations

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# Learning Objectives (1 of 2)

- Illustrate the importance that project schedules and good project schedule management can have in helping to make projects successful
- Discuss the process of planning schedule management
- Define activities as the basis for developing project schedules
- Describe how project managers use network diagrams and dependencies to assist in activity sequencing
- Explain how various tools and techniques help project managers perform activity duration estimates
- Use a Gantt chart for planning and tracking schedule information, find the critical path for a project, and describe how the Program Evaluation and Review Technique (PERT) and critical chain scheduling affect schedule development

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# Learning Objectives (2 of 2)

- Discuss how discipline are involved in controlling and managing changes to the project schedule
- Describe how project management software can assist in project schedule management and review words of caution before using this software
- Discuss considerations for agile/adaptive environments

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# The Importance of Project Schedules

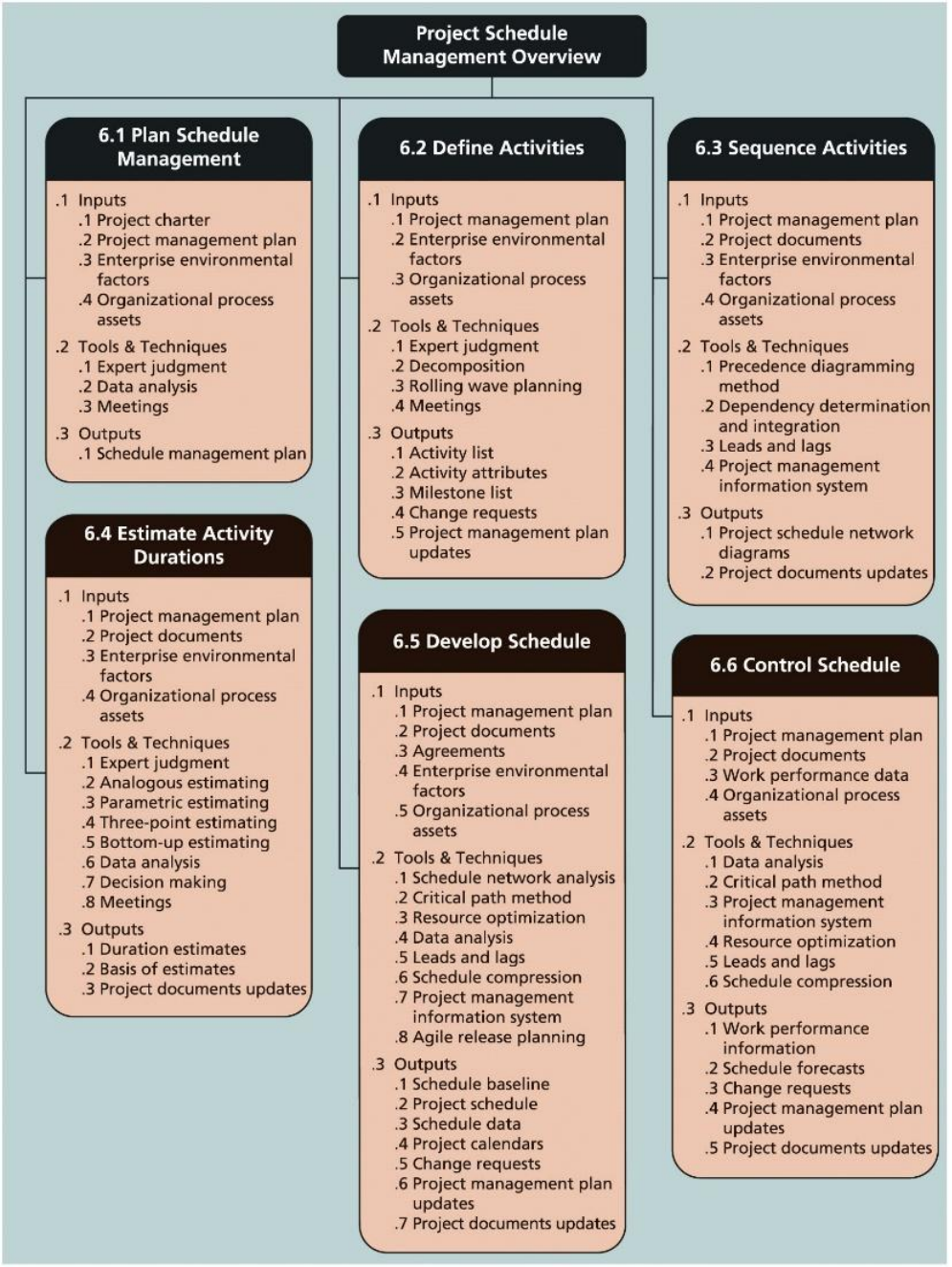
- Project managers often cite delivering projects on time as one of their biggest challenges
  - Time has the least amount of flexibility; it passes no matter what happens on a project
- Individual work styles and cultural differences may also cause schedule conflicts
  - Working hours
  - Holidays

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# Project Schedule Management Processes

1. Planning schedule management
2. Defining activities
3. Sequencing activities
4. Estimating activity durations
5. Developing the schedule
6. Controlling the schedule

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**FIGURE 6-1** Project schedule management overview

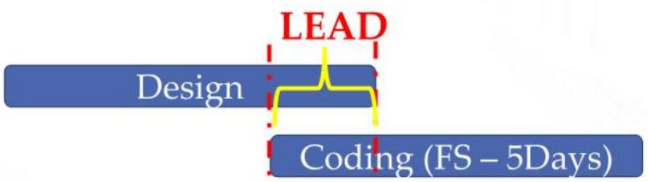
# 1. Planning Schedule Management

- Elements of a schedule management plan
  - Project schedule model development
    - Model of activities, dependencies, estimated durations, resources but without calendar dates assigned.
    - I.e project schedule = schedule model + calendar dates (to start running the project)
  - Level of accuracy and units of measure
    - How accurate schedule estimates should be and time unit used (e.g. hours, days)
  - Control thresholds
    - Variance threshold for monitoring schedule performance, e.g. ± 10%
  - Rules of performance measurement
    - How team members determine the percentage of work completed
  - Reporting formats
    - Format and frequency of schedule reports
  - Process descriptions
    - How all schedule management processes will be performed

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# 2. Defining Activities (1 of 2)

- Defining activities involves identifying the specific actions that will produce the project deliverables in enough detail to determine resource and schedule estimates
  - Activities are driven by deliverables on WBS
  - Activity list: a tabulation of activities to be included on a project schedule
    - Activity name, activity identifier or number, and brief description of the activity
  - Activity attributes provide schedule-related information
    - Predecessors, successors, logical relationships, leads and lags, resource requirements, constraints, imposed dates, and assumptions related to the activity
      - (Lead = a relationship whereby the successor activity begins before the predecessor activity has completed
      - Lag = a relationship whereby the successor activity cannot start right after the end of its predecessor)







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https://blog.masterofproject.com/leads-and-lags/

# 2. Defining Activities (2 of 2)

- Milestone is a significant event that normally has no duration
  - It often takes several activities and a lot of work to complete a milestone
  - They are useful tools for setting schedule goals and monitoring progress
    - Examples: obtaining customer sign-off on key documents, completion of specific products
- Most project management software use the term task for activity

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# 3. Sequencing Activities (1 of 5)

- Sequencing process involves evaluating the reasons for dependencies and the different types of dependencies
- A dependency or relationship is the sequencing of project activities or tasks
  - Mandatory dependencies
    - Inherent in the nature of the work being performed on a project, sometimes referred to as hard logic, e.g. testing must be after coding is finished
  - Discretionary dependencies
    - Defined by the project team, sometimes referred to as soft logic, and should be used with care since they may limit later scheduling options, e.g. detailed design must be after user sign-off on analysis work

    vs.
  - External dependencies
    - Involve relationships between project and non-project activities, e.g. software installation must be after delivery of new hardware from a supplier
  - Internal dependencies
    - Involve relationships between project activities inside project team's control, e.g. system testing is after unit testing

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# Sequencing Activities (2 of 5)

- Network diagrams are the preferred technique for showing activity sequencing
  - Schematic display of the logical relationships among, or sequencing of, project activities
  - Not every items on WBS needs to be shown; only activities with dependencies need to be shown
  - Some people put start and end milestones and list all activities, while some put only summary tasks
  - Two main formats are the arrow diagramming method (ADM) and precedence diagramming method (PDM)

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# Sequencing Activities (3 of 5)

- Arrow diagramming method (ADM) (i.e. activity-on-arrow (AOA) network diagrams)
  - Activities are represented by arrows
  - Nodes or circles are the starting and ending points of activities
  - Only show finish-to-start dependencies

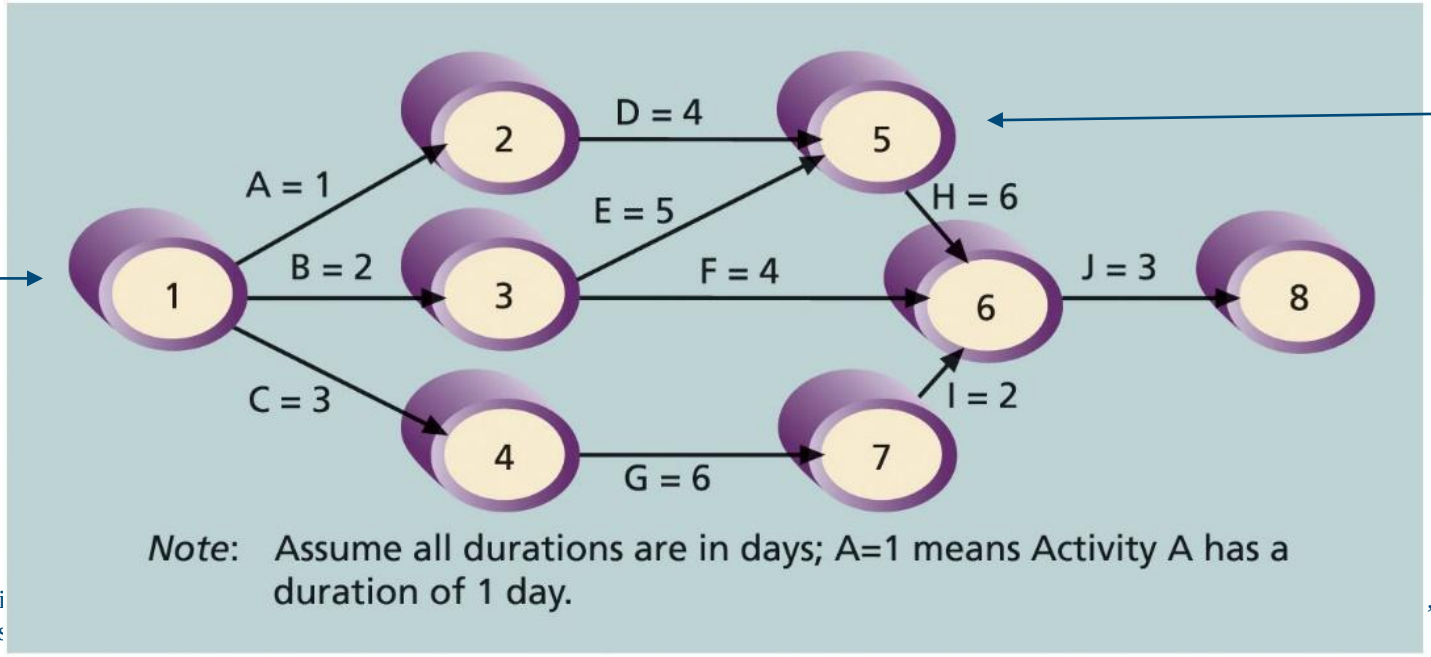



FIGURE 6-2 Network diagram for project X

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# Sequencing Activities (4 of 5)

- Precedence diagramming method (PDM) (i.e. activity-on-node (AON) network diagrams)
  - Network diagramming technique in which boxes represent activities

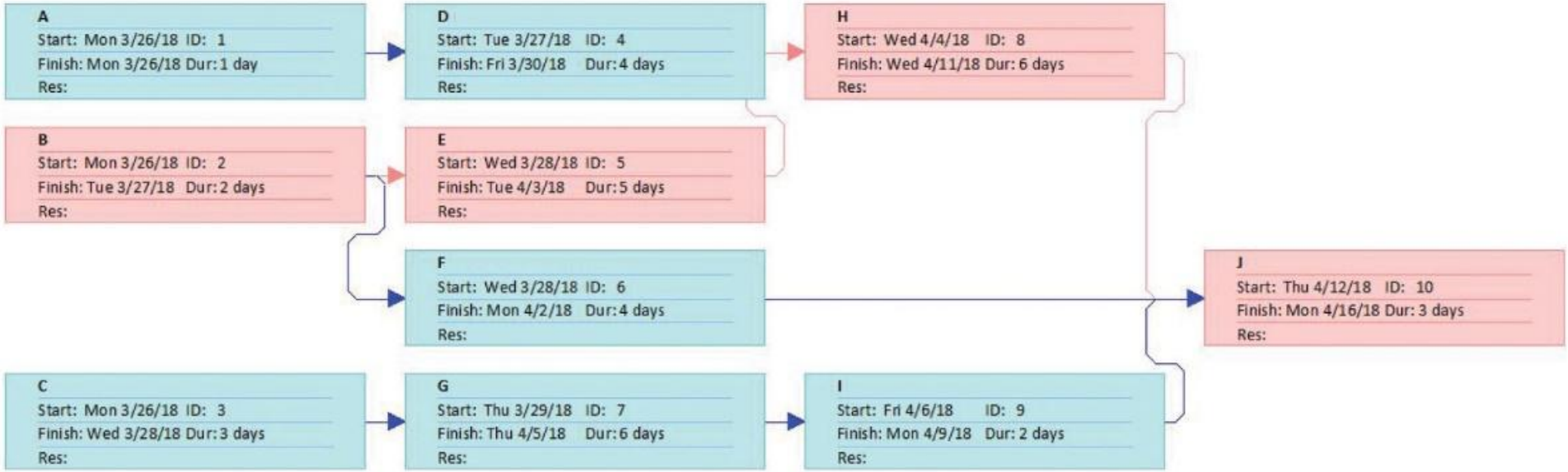



**FIGURE 6-4** Precedence diagramming methods (PDM) network diagram for project X

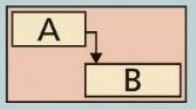
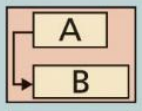
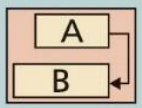
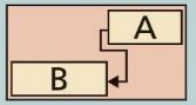
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# Sequencing Activities (5 of 5)

- Types of dependencies or relationships between activities in PDM

**Task dependencies**

The nature of the relationship between two linked tasks. You link tasks by defining a dependency between their finish and start dates. For example, the "Contact caterers" task must finish before the start of the "Determine menus" task. There are four kinds of task dependencies in Microsoft Project.

| Task dependency | Example | Description |
|---|---|---|
| Finish-to-start (FS) |   | Task (B) cannot start until task (A) finishes. |
| Start-to-start (SS) |   | Task (B) cannot start until task (A) starts. Or Must-start-together relationship |
| Finish-to-finish (FF) |   | Task (B) cannot finish until task (A) finishes. |
| Start-to-finish (SF) |   | Task (B) cannot finish until task (A) starts. |

A – Contact caterers, B – Determine menus
A – Test system, B – Install system

A – Pour asphalt, B – Level road
A – Write code, B – Review code

A – Lay wires, B – Inspect wires
A – Write code for module X, B – Unit test module X

A – Support take-over by operation, B - Support by project team after go-live

**FIGURE 6-3** Task dependency types

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# 4. Estimating Activity Durations (1 of 3)

- Duration includes the actual amount of time worked on an activity *plus* elapsed time
  - Effort is the number of workdays or work hours required to complete a task and does not normally equal duration
  - E.g 20 hours of work (effort) may be spread out over 2 months (duration)
- People doing the work should help create estimates
  - An expert should review the estimates
  - It is helpful to review similar projects
  - Consider availability of resources especially human resources
  - Update the estimates as the project progresses, e.g. scope changes
- A three-point estimate is an estimate that includes an optimistic, most likely, and pessimistic estimate
  - Three-point estimates are needed for PERT and Monte Carlo simulations

$$average(optimistic\ time, most\ likely\ time, pessimistic\ time)$$

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# Estimating Activity Durations (2 of 3)

WBS by phase (i.e. process decomposition) in tabular form of a Scrum project (Ride Sharing)

Activity (Task)

Process decomposition

| Task Number | Task Name | Duration (days) | Dependency | Status |
|---|---|---|---|---|
| 1 | Initiating | | | |
| 1.1 | Develop a Business Case | 5 | | Complete |
| 1.2 | Undertake a Feasibility Study | 2 | | Complete |
| 1.3 | Establish the Project Charter | 1 | 1.2 | Complete |
| 1.4 | Create the Project Team and Determine Roles | 1 | 1.3 | Complete |
| 1.5 | Determine Total Number of Sprints and Releases | 3 | 1.4 | Complete |
| 1.6 | Kick-Off Meeting | 1 | 1.5 | Complete |
| 2 | Planning | | | |
| 2.1 | Create Product Backlog and Sprint Backlog | 7 | 1.6 | Open |
| 2.2 | Create Document for Sprint Planning | 5 | 2.1 | Open |
| 2.3 | Create a Resource Plan | 3 | 2.2 | Open |
| 2.4 | Create a Financial Plan | 5 | 2.3 | Open |
| 2.5 | Create a Budget Plan | 3 | 2.4 | Open |
| 3 | Development | | | |
| 3.1 | Requirement Gathering | | | |
| 3.1.1 | Identify User's need and Survey | 5 | 2.5 | Open |
| 3.2 | Training | | | |
| 3.2.1 | Knowledge Sharing Session | 2 | 3.1.1 | Open |
| 3.2.2 | Technology and Tools Session | 2 | 3.2.1 | Open |
| 3.2.3 | Setup Session | 4 | 3.2.2 | Open |
| 3.2.4 | Existing System Session | 2 | 3.2.3 | Open |
| 3.3 | Executing (System & Testing) | | | |
| 3.3.1 | Sprint 1 | | | |
| 3.3.1.1 | Sprint Planning Route System, Meter Calculator System | 1 | 3.2.4 | Open |
| 3.3.1.2 | Daily Scrums | 14 | 3.2.4 | Open |
| 3.3.1.3 | Implement | 11 | 3.3.1.1 | Open |
| 3.3.1.4 | Testing | 8 | 3.3.1.3 | Open |
| 3.3.1.5 | Sprint Review 1 | 1 | 3.3.1.4 | Open |
| 3.3.1.6 | Sprint Retrospective 1 | 1 | 3.3.1.5 | Open |
| 3.3.2 | Sprint 2 | | | |
| 3.3.2.1 | Sprint Planning Accounting Management System | 1 | 3.3.1.6 | Open |
| 3.3.2.2 | Daily Scrums | 14 | 3.3.1.6 | Open |
| 3.3.2.3 | Implement | 11 | 3.3.2.1 | Open |
| 3.3.2.4 | Testing | 8 | 3.3.2.3 | Open |
| 3.3.2.5 | Sprint Review 2 | 1 | 3.3.2.4 | Open |
| 3.3.2.6 | Sprint Retrospective 2 | 1 | 3.3.2.5 | Open |
| 3.3.3 | Sprint 3 | | | |
| 3.3.3.1 | Sprint Planning Messaging System | 1 | 3.3.2.6 | Open |
| 3.3.3.2 | Daily Scrums | 14 | 3.3.2.6 | Open |
| 3.3.3.3 | Implement | 11 | 3.3.3.1 | Open |
| 3.3.3.4 | Testing | 8 | 3.3.3.3 | Open |
| 3.3.3.5 | Sprint Review 3 | 1 | 3.3.3.4 | Open |
| 3.3.3.6 | Sprint Retrospective 3 | 1 | 3.3.3.5 | Open |
| 3.3.4 | Sprint 4 | | | |
| 3.3.4.1 | Sprint Planning Route Notification, User Notification | 1 | 3.3.3.6 | Open |
| 3.3.4.2 | Daily Scrums | 14 | 3.3.3.6 | Open |
| 3.3.4.3 | Implement | 11 | 3.3.4.1 | Open |
| 3.3.4.4 | Testing | 8 | 3.3.4.3 | Open |
| 3.3.4.5 | Sprint Review 4 | 1 | 3.3.4.4 | Open |
| 3.3.4.6 | Sprint Retrospective 4 | 1 | 3.3.4.5 | Open |
| 3.3.5 | Software Release | 2 | 3.3.4.6 | Open |
| 3.4 | Documentation | | | |
| 3.4.1 | User's Guide | 4 | 3.3.5 | Open |
| 3.4.2 | Installation Guide | 1 | 3.4.1 | Open |
| 4 | Monitoring and Controlling | | | |
| 4.1 | Perform Burndown Chart | 71 | | Open |
| 4.2 | Perform Cost Management | 71 | | Open |
| 4.3 | Perform Risk Management | 71 | | Open |
| 4.4 | Perform Change Management | 71 | | Open |
| 5 | Closing | | | |
| 5.1 | Review Project Completion | 3 | 3.4.2 | Open |

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# Estimating Activity Durations (3 of 3)

Zoom in on planning sprint backlog

Activities (tasks) with duration (effort) but no dependency

Estimate effort (person-hours)

| | EPIC ID | EPIC Name | User Story ID | User Story | Task | Volunteer | | Sta | Esti |
|---|---|---|---|---|---|---|---|---|---|
| Sprint1 | EPIC1 | Matching system | US1-1 | As a patient<br>I want to select a physician<br>so that I can consult a physician | Setup backend server & | Atiwat | Tarm | Done | 4 |
| | | | | | Setup Database (mongodb) | Tarm | Atiwat | Done | 2 |
| | | | | | Setup React, folder | Kittipong | Kanokpich | Done | 3 |
| | | | | | Setup tailwindcss and design | Kittipong | Jirawat | Done | 2 |
| | | | | | Design model schema | Tarm | Thitaree | Done | 8 |
| | | | | | Implement API for fetching | Tarm | Atiwat | Done | 1 |
| | | | | | Implement API for matching | Thitaree | Atiwat | Done | 2 |
| | | | | | Implement API for notification | Thitaree | Atiwat | Done | 2 |
| | | | | | Implement physician data | Jirawat | Nopdanai | Done | 3 |
| | | | | | Implement comment card | Nopdanai | Raviporn | Done | 2 |
| | | | | | Implement notification button | Raviporn | Nopdanai | Done | 5 |
| | | | | | Implement other buttons (The | Kanokpich | Raviporn | Done | 2 |
| | | | | | Implement patient's search | Kittipong | Raviporn | Done | 2 |
| | | | | | Implement physician | Nopdanai | Kanokpich | Done | 2 |
| | | | | | Implement after selection | Jirawat | Raviporn | Done | 1 |
| | | | | | Implement physician page | Kanokpich | Kittipong | Done | 2 |
| | | | US1-2 | As a physician<br>I want to accept patients' request to consult with me<br>so that I can give consultations to patients | Implement patient | Jirawat | Nopdanai | Done | 3 |
| | | | | | Finishing touch on implementing the physician page | Kanokpich | Kittipong | Done | 2 |
| | | | | **AND** | Implement patient detail page | Jirawat | Nopdanai | Done | 2 |
| | | | US1-3 | As a physician | Implement an update-request | Atiwat | Thitaree | Done | 2 |
| | | | US1-4 | As a patient<br>I want to be able to sort the physician's list by the relevant | Implement sorted icon and | Jirawat | Kanokpich | Done | 3 |
| | | | | | Implement http method for | Kittipong | Nopdanai | Done | 3 |
| | | | | | modify fetching-physician- | Atiwat | Tarm | Done | 2 |
| | | | US1-5 | As a patient<br>I want to see the physician's working hours | Design physician information | Raviporn | Kanokpich | Done | 1 |
| | | | | | Implement physician | Kanokpich | Nopdanai | Done | 5 |
| | | | | | Implement a get-working-hour | Thitaree | Atiwat | Done | 1 |
| | | | US1-6 | As a patient<br>I want to search physicians<br>so that I can select appropriate | Implement search box | Kanokpich | Jirawat | Done | 2 |
| | | | | | Implement http method for | Kittipong | Nopdanai | Done | 4 |
| | | | | | Implement search-physician | Atiwat | Tarm | Done | 3 |
| | | | US1-7 | As a patient<br>I want to see the matching status<br>so that I can know where my request state is in | Design consultation page | Kanokpich | Raviporn | Done | 2 |
| | | | | | Design matching status card | Raviporn | Thitaree | Done | 1 |
| | | | | | Implement consultation page | Jirawat | Kanokpich | Done | 3 |
| | | | | | Implement menu (hamburger) | Raviporn | Kittipong | Done | 4 |
| | | | | | Implement get-matching- | Tarm | Thitaree | Done | 2 |
| | | | US1-8 | As a patient<br>I want to cancel the request | Implement http method for | Kanokpich | Jirawat | Done | 1 |
| | | | | | add cancel request in update- | Atiwat | Thitaree | Done | 2 |
| | EPIC2 | Registration and Login system | US2-1 | As a patient<br>I want to register to the system<br>so that I can consult a physician | Implement patient-register | Atiwat | Thitaree | Done | 2 |
| | | | | | Implement custom textbox | Kittipong | Raviporn | Done | 3 |
| | | | | | Implement patient's | Jirawat | Nopdanai | Done | 3 |
| | | | US2-2 | As a physician<br>I want to register to the system | Implement physician-register | Thitaree | Tarm | Done | 2 |
| | | | | | Implement physician | Kanokpich | Jirawat | Done | 2 |
| | | | US2-3 | As a patient<br>I want to provide a credit/debit card information<br>so that I can pay service fees | Design card addition page | Raviporn | Thitaree | Done | 1 |
| | | | | | Implement card addition page | Raviporn | Jirawat | Done | 2 |
| | | | | | Design credit card schema | Thitaree | Atiwat | Done | 1 |
| | | | | | Implement add-credit-card | Atiwat | Tarm | Done | 2 |
| | | | US2-4 | As a physician<br>I want to provide my Medical | Implement physician | Raviporn | Kittipong | Done | 2 |
| | | | | | modify physician-register | Atiwat | Thitaree | Done | 1 |

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# 5. Developing the Schedule

- Uses results of the other schedule management processes to determine the start and end date of the project
  - Ultimate goal is to create a realistic project schedule that provides a basis for monitoring project progress for the time dimension of the project
- Important tools and techniques
  - Gantt charts
  - Critical path analysis
  - PERT analysis
  - Critical chain scheduling

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# Gantt Charts (1 of 7)

- Provide a standard format for displaying project schedule information by listing project activities and corresponding start and finish dates in a calendar form
  - Symbols
    - Diamond: milestones
    - Thick bars: summary tasks
    - Light horizontal bars: durations of tasks
    - Arrows: dependencies between tasks

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# Gantt Charts (2 of 7)

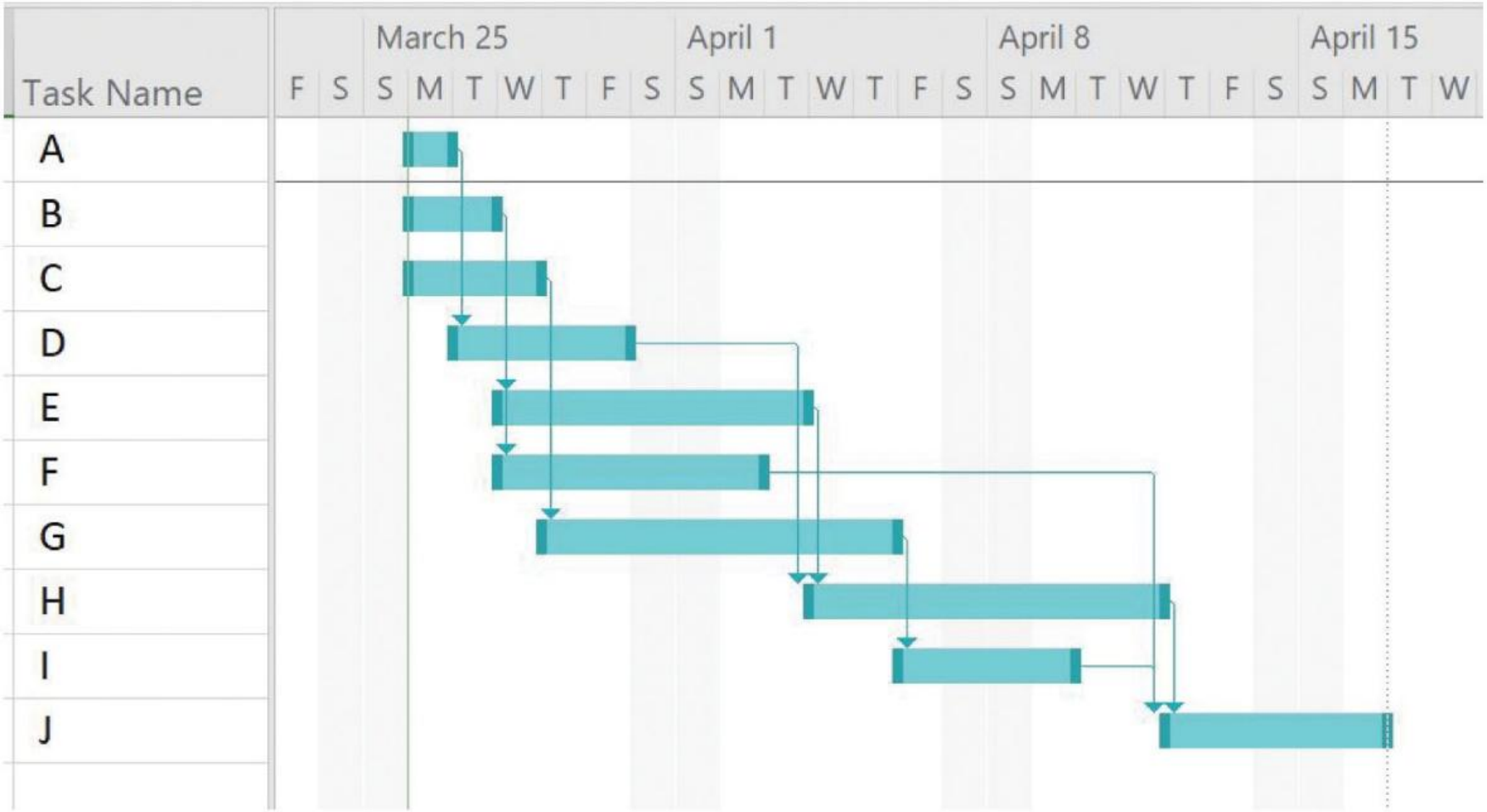



**FIGURE 6-5** Gantt chart for project X (slide 12, 13)

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# Gantt Charts (3 of 7)

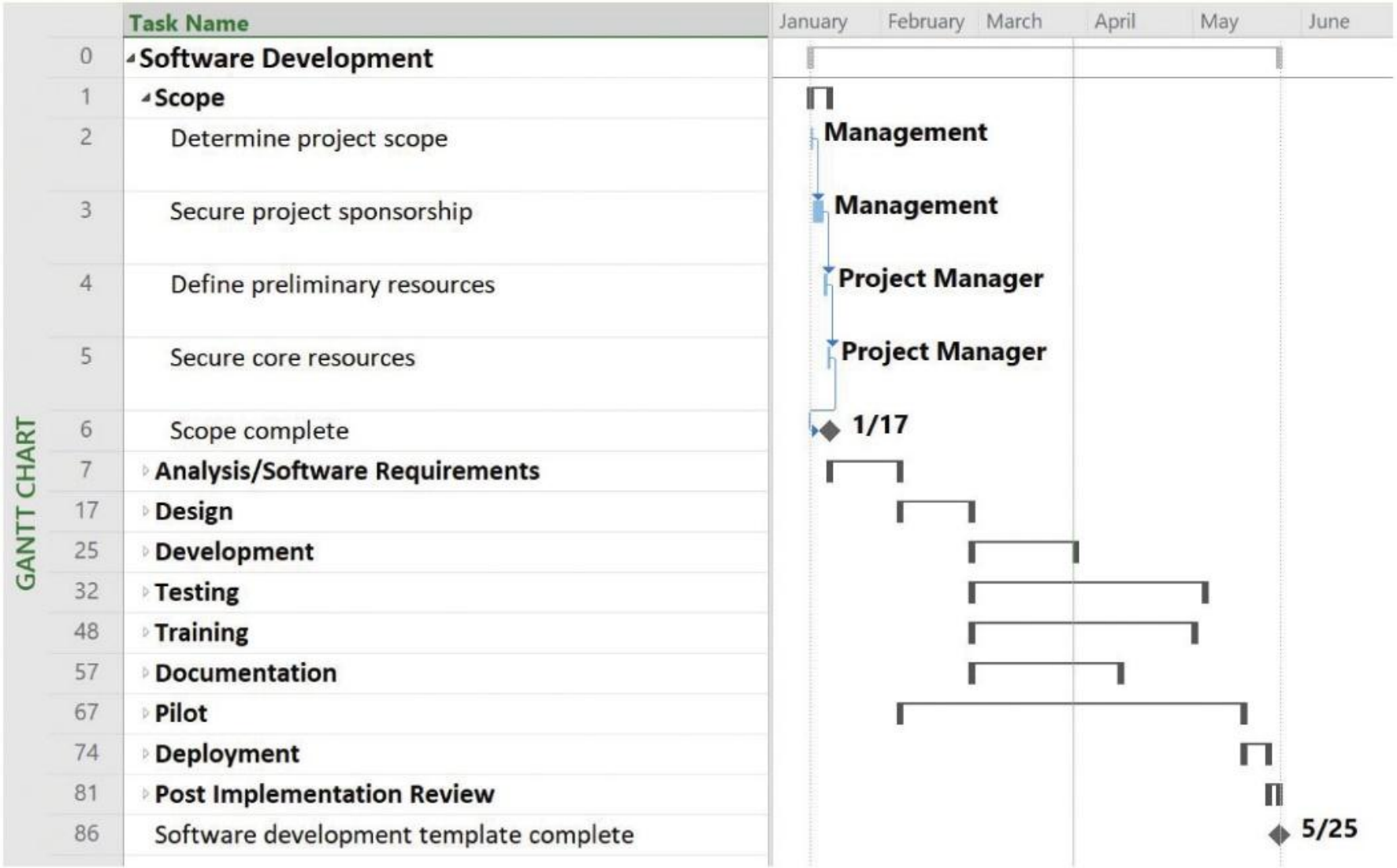



**FIGURE 6-6** Gantt chart for software launch project

# Gantt Charts (4 of 7)

- Adding milestones to Gantt charts
  - Many people like to focus on meeting milestones, especially for large projects
    - Milestones emphasize important events or accomplishments on projects
- SMART Criteria for milestones
  - Specific – clear and target a specific area
  - Measurable – measurable that the milstone is achieved
  - Assignable – responsible person
  - Realistic – achievable event
  - Time-framed – scheduled at appropriate time

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# Gantt Charts (5 of 7)

- Tracking Gantt charts
  - A special form of Gantt chart that shows progress on a project by comparing planned and actual schedule information
  - Baseline dates – planned schedule dates for activities
  - Schedule baseline – entire approved planned schedule
- It has columns Start, Finish, Baseline Start, Baseline Finish
- It is based on the percentage of work completed for project tasks or the actual start and finish dates

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# Gantt Charts (6 of 7)

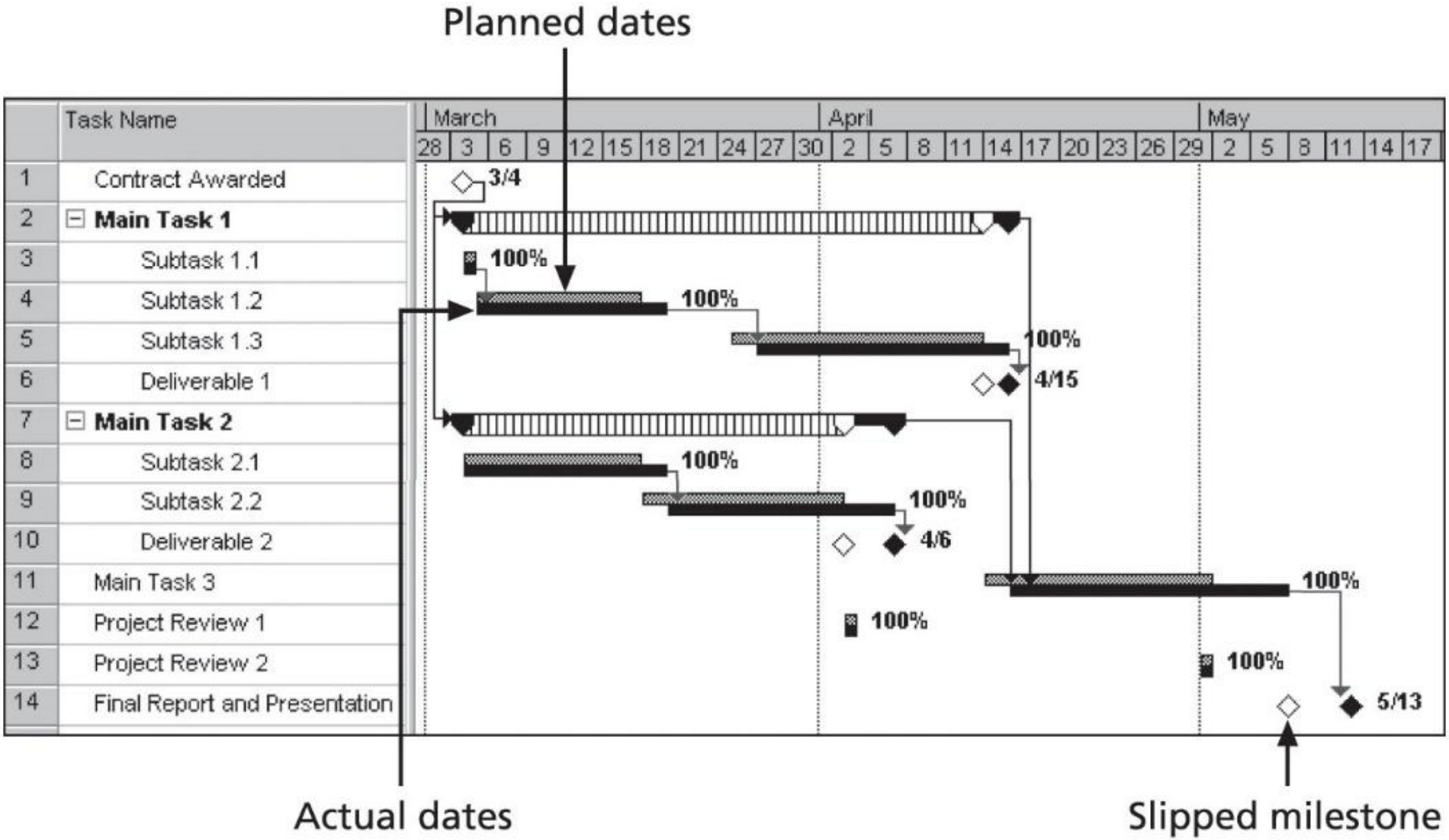
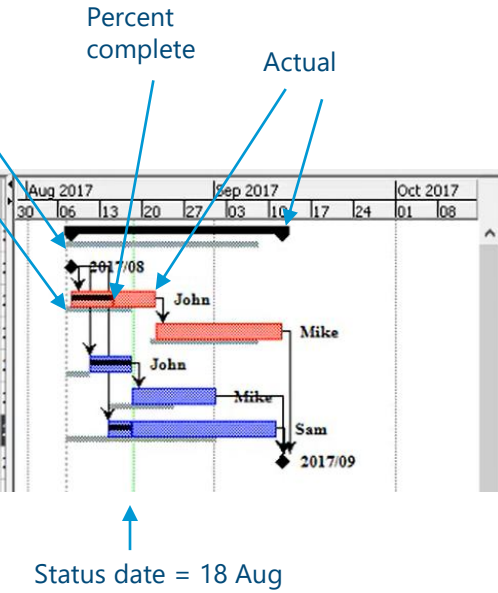



**FIGURE 6-7** Sample tracking Gantt chart

# Gantt Charts (7 of 7)

| | | Name | Percent Comp... | Duration | Baseline Start | Actual Start | Baseline Finish | Actual Finish |
|---|---|---|---|---|---|---|---|---|
| 1 | | Tracking Project Progres | 23% | 25 days | 2017/08/07 8:00 AM | 2017/08/08 10:00 AM | 2017/09/08 5:00 PM | |
| 2 | | Start | 0% | 0 days | 2017/08/07 8:00 AM | | 2017/08/07 8:00 AM | |
| 3 | | Tasl A | 50% | 10 days | 2017/08/07 8:00 AM | 2017/08/08 10:00 AM | 2017/08/18 5:00 PM | |
| 4 | | Task B | 0% | 15 days | 2017/08/21 8:00 AM | | 2017/09/08 5:00 PM | |
| 5 | ✔ | Task C | 100% | 5 days | 2017/08/07 8:00 AM | 2017/08/11 10:00 AM | 2017/08/11 5:00 PM | 2017/08/18 10:00 AM |
| 6 | | Task D | 0% | 10 days | 2017/08/14 8:00 AM | | 2017/08/25 5:00 PM | |
| 7 | | Task E | 20% | 20 days | 2017/08/07 8:00 AM | 2017/08/14 10:00 AM | 2017/09/01 5:00 PM | |
| 8 | | Finish | 0% | 0 days | 2017/09/08 5:00 PM | | 2017/09/08 5:00 PM | |




https://www.youtube.com/watch?v=lJyaEWRBT6Y

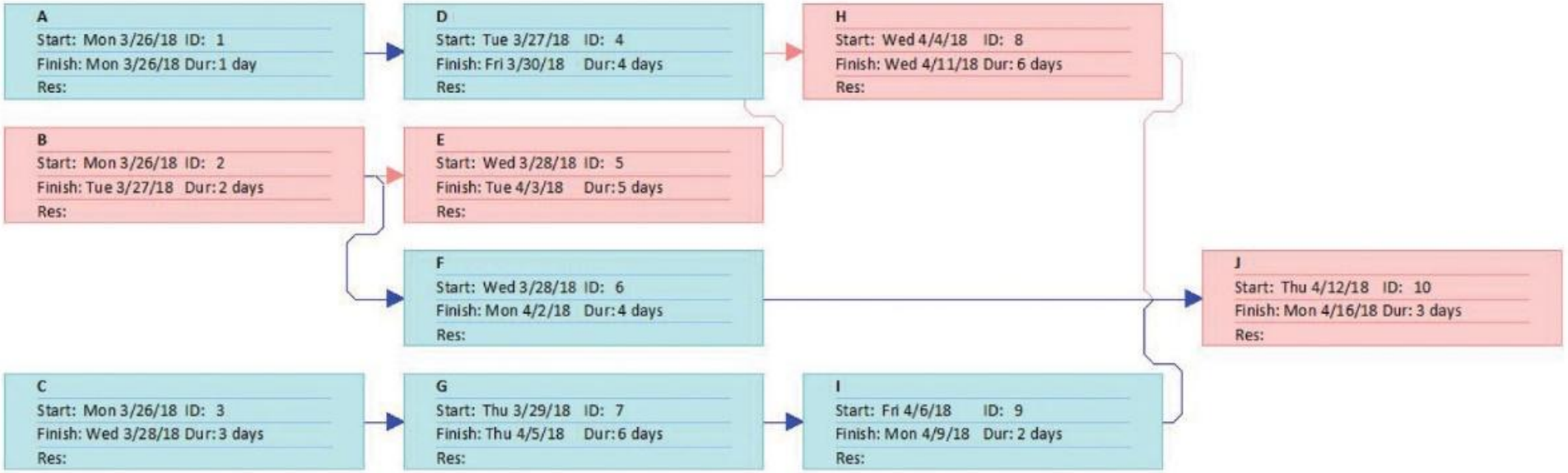
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# Critical Path Method (CPM) (1 of 2)

- Network diagramming technique used to predict total project duration
  - Critical path: series of activities that determine the earliest time by which the project can be completed
    - The longest path through the network diagram and has the least amount of slack or float
    - Slack (or float) is amount of time an activity may be delayed without delaying a succeeding activity or the project finish date
- Calculating the critical path
  - Develop a good network diagram and add the duration estimates for all activities on each path through the network diagram
    - Longest path is the critical path
  - If one or more of the activities on the critical path (i.e. critical activities) takes longer than planned, the whole project schedule will slip unless the project manager takes corrective action

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# Critical Path Method (CPM) (2 of 2)




*Note*: Assume all durations are in days.

| | | |
|---|---|---|
| Path 1: | A-D-H-J | Length = 1+4+6+3 = 14 days |
| Path 2: | B-E-H-J | Length = 2+5+6+3 = 16 days |
| Path 3: | B-F-J | Length = 2+4+3 = 9 days |
| Path 4: | C-G-I-J | Length = 3+6+2+3 = 14 days |

Since the critical path is the longest path through the network diagram, Path 2, B-E-H-J, is the critical path for Project X.

**FIGURE 6-8** Determining the critical path for project X (slide 13)

# Growing Grass Can Be on the Critical Path

- The fact that the term critical path includes the word critical does not mean that it includes all "critical" activities
  - Only accounts for time
    - Example: growing grass for Disney's Animal Kingdom
- There can be more than one critical path if the lengths of two or more paths are the same
  - Project managers should closely monitor performance of activities on the critical path to avoid late project completion
- Critical path can change as the project progresses

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# Program Evaluation and Review Technique (PERT)

- Network analysis technique used to estimate project duration when there is a high degree of uncertainty about the individual activity duration estimates
  - Uses arrow diagramming method (ADM) (i.e. activity-on-arrow (AOA) network diagrams
  - Uses probabilistic time estimates, applying three-point estimates of activity durations
  - By using the PERT weighted average for each activity duration estimate (O : M : P = 1 : 4 : 1), total project duration estimate takes into account the risk or uncertainty in the individual activity estimates

$$PERT\ weighted\ average = \frac{optimistic\ time + 4 * most\ likely\ time + pessimistic\ time}{6}$$

  - Applies to the critical path method (CPM) as PERT/CPM

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# Using Critical Path Analysis to Make Schedule Trade-Offs (1 of 5)

- If a task on the critical path is behind schedule, the project manager must be aware and decide what to do
- Early start (ES) – earliest time an activity can start based on project network logic
- Early finish (EF) – earliest time an activity can finish based on project network logic
- Late start (LS) – latest time an activity can start without delaying the project finish date
- Late finish (LF) – latest time an activity can finish without delaying the project finish date
- Forward pass – process to determine the early start and early finish for each activity
- Backward pass - process to determine the late start and late finish for each activity
- Free slack (FS) or free float (FF) - amount of time an activity can be delayed from its early start without delaying the early start of succeeding activities
- Total slack (TS) or total float (TF) - amount of time an activity can be delayed from its early start without delaying the planned project finish date

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# Using Critical Path Analysis to Make Schedule Trade-Offs (2 of 5)

- Early start (ES) and Early finish (EF) (in forward pass)
  - ES of first activity = 0
  - EF = ES + dur
  - ES of succeeding activity = max EF of preceding activities
- Late start (LS) and Late finish (LF) (in backward pass)
  - LF of last activity = EF of last activity
  - LS = LF – dur
  - LF of preceding activity = min LS of succeeding activities

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# Using Critical Path Analysis to Make Schedule Trade-Offs (3 of 5)

- Total slack (TS)
  - TS = LS – ES or LF – EF
    - If the start of an activity is delayed but still within its total slack time, no effect on project finish date
  - TS of critical activity = 0
    - If the start of a critical activity is delayed, consider reallocating resources from activities with total slack > 0
- Free slack (FS)
  - FS = (min ES of succeeding activities) – ES – dur
- FS ≤ TS
  - FS of critical activity = 0



How to calculate total slack https://www.youtube.com/watch?v=TQcJ4bJDhQQ

What is free slack (free float)
https://www.youtube.com/watch?v=ErsolVw9UKE&t=0s&index=5&list=PLP_0l6U5zhU6FZojgg_0G5SeRAVY9Ocvu

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# Using Critical Path Analysis to Make Schedule Trade-Offs (4 of 5)

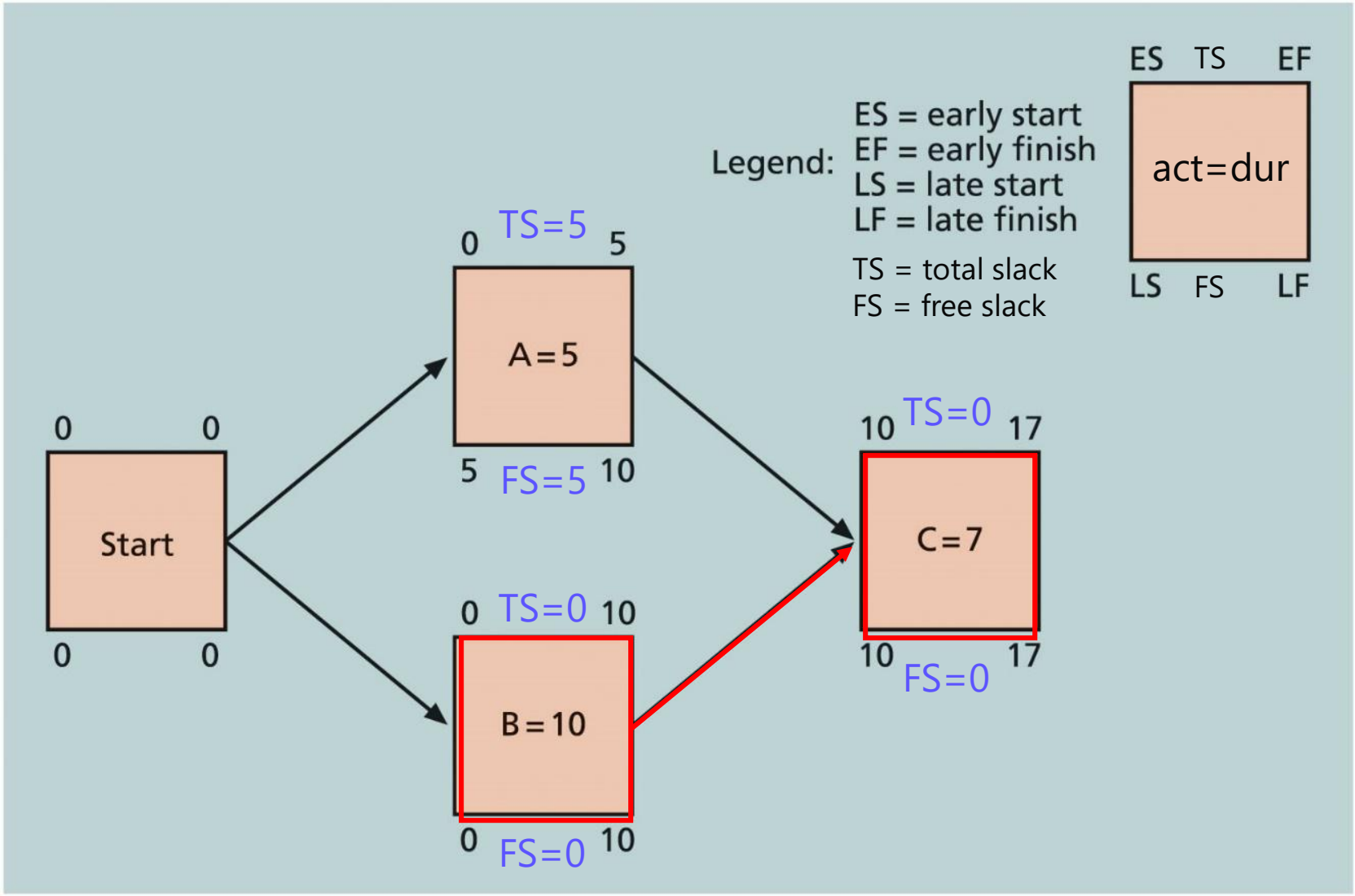



**FIGURE 6-9** Calculating early and late start and finish dates

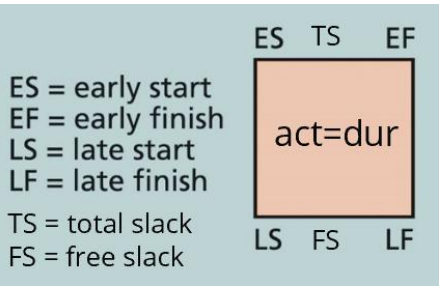
# Using Critical Path Analysis to Make Schedule Trade-Offs (5 of 5)

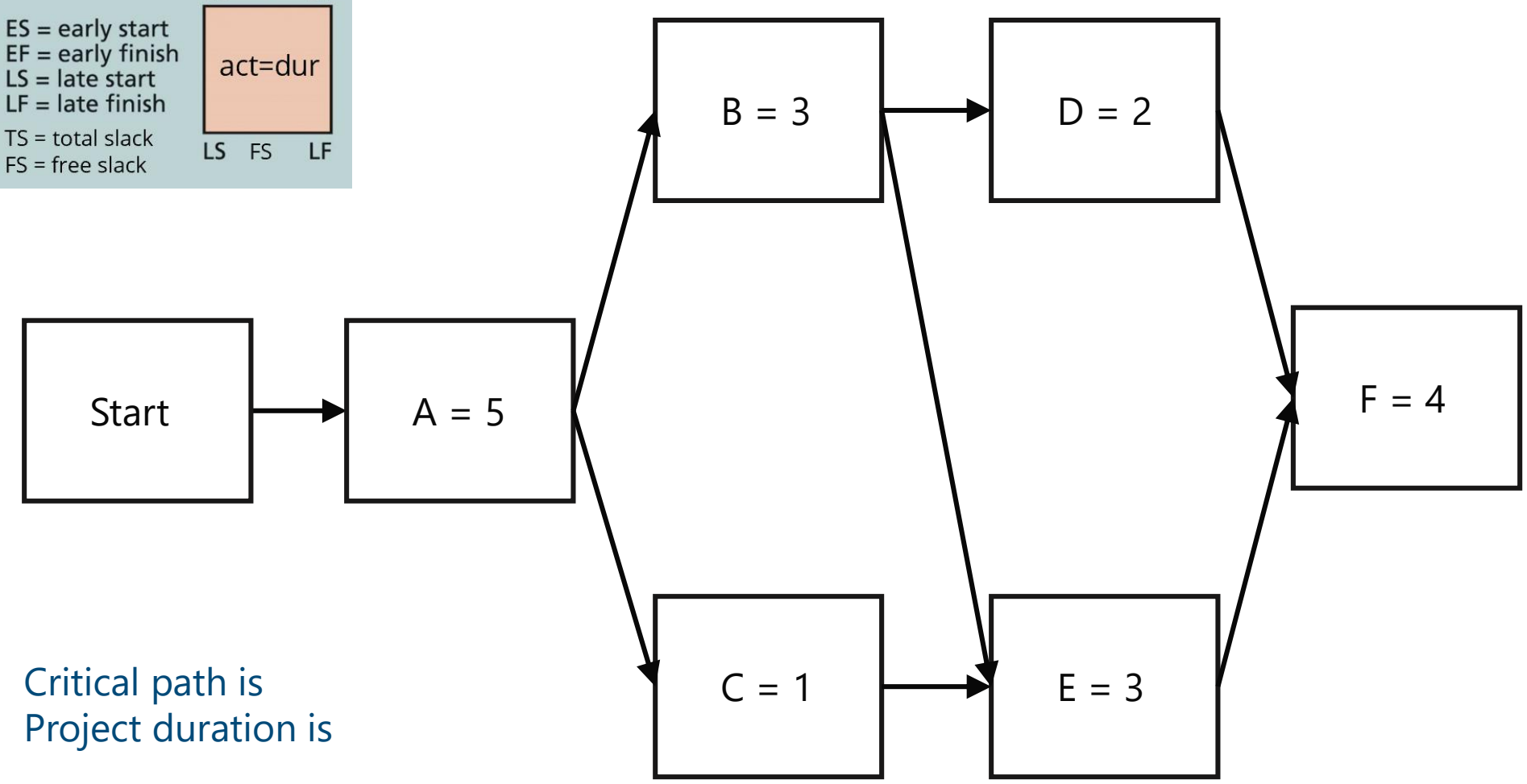
| Task Name | Start | Finish | Late Start | Late Finish | Free Slack | Total Slack |
|---|---|---|---|---|---|---|
| A | 8/3/15 | 8/3/15 | 8/5/15 | 8/5/15 | 0d | 2d |
| B | 8/3/15 | 8/4/15 | 8/3/15 | 8/4/15 | 0d | 0d |
| C | 8/3/15 | 8/5/15 | 8/5/15 | 8/7/15 | 0d | 2d |
| D | 8/4/15 | 8/7/15 | 8/6/15 | 8/11/15 | 2d | 2d |
| E | 8/5/15 | 8/11/15 | 8/5/15 | 8/11/15 | 0d | 0d |
| F | 8/5/15 | 8/10/15 | 8/14/15 | 8/17/15 | 7d | 7d |
| G | 8/6/15 | 8/13/15 | 8/10/15 | 8/17/15 | 0d | 2d |
| H | 8/12/15 | 8/19/15 | 8/12/15 | 8/19/15 | 0d | 0d |
| I | 8/14/15 | 8/17/15 | 8/18/15 | 8/19/15 | 2d | 2d |
| J | 8/20/15 | 8/24/15 | 8/20/15 | 8/24/15 | 0d | 0d |

Table 6-1 Free and Total Float or Slack for Project X (slide 12,13)

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# What is the critical path and project duration? What are ES, EF, LS, LF, TS, FS of these activities?



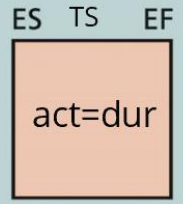




Critical path is

Project duration is

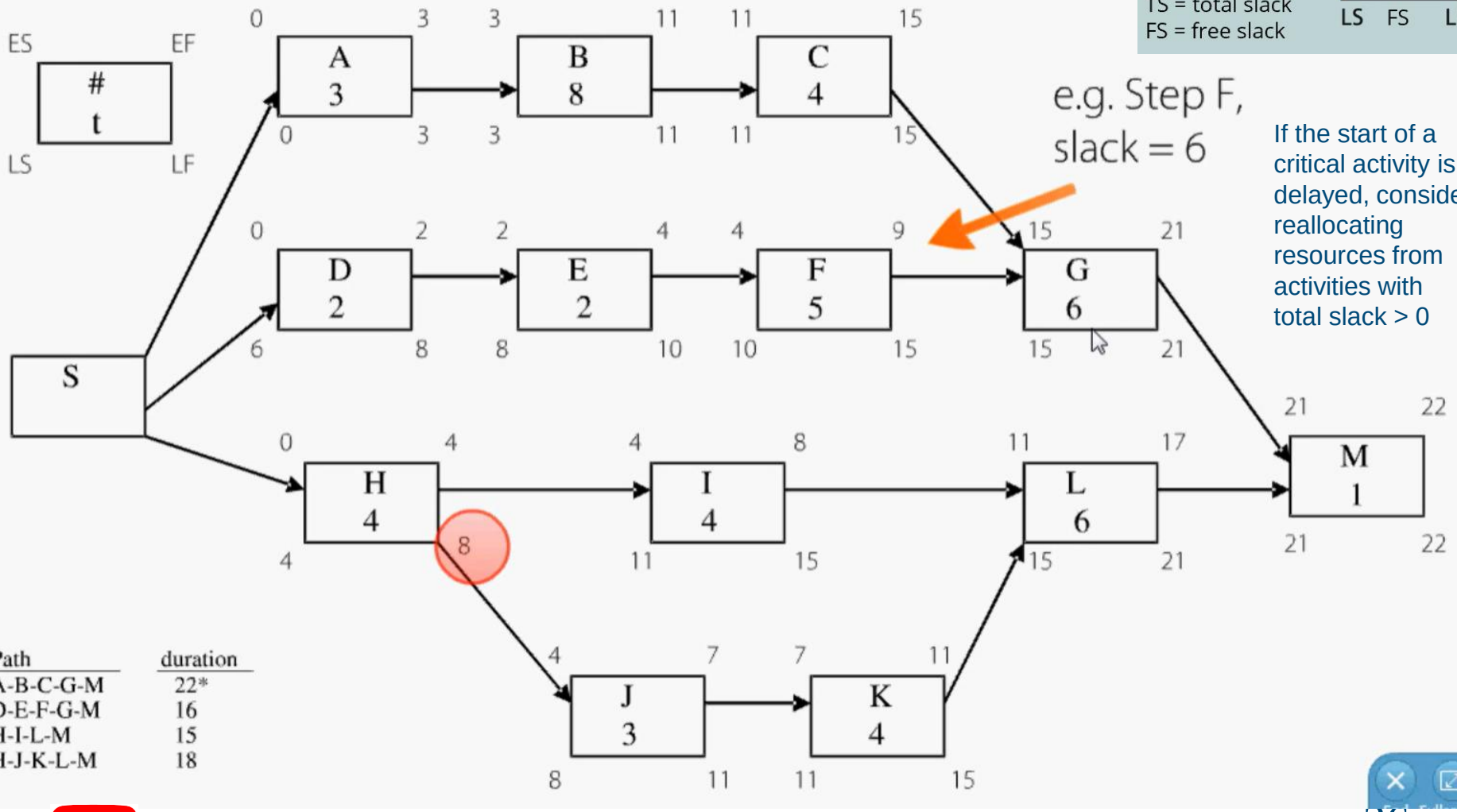
# What is the critical path and project duration? What are ES, EF, LS, LF, TS, FS of these activities?

ES = early start
EF = early finish
LS = late start
LF = late finish
TS = total slack
FS = free slack










| Path | duration |
|---|---|
| A-B-C-G-M | 22* |
| D-E-F-G-M | 16 |
| H-I-L-M | 15 |
| H-J-K-L-M | 18 |

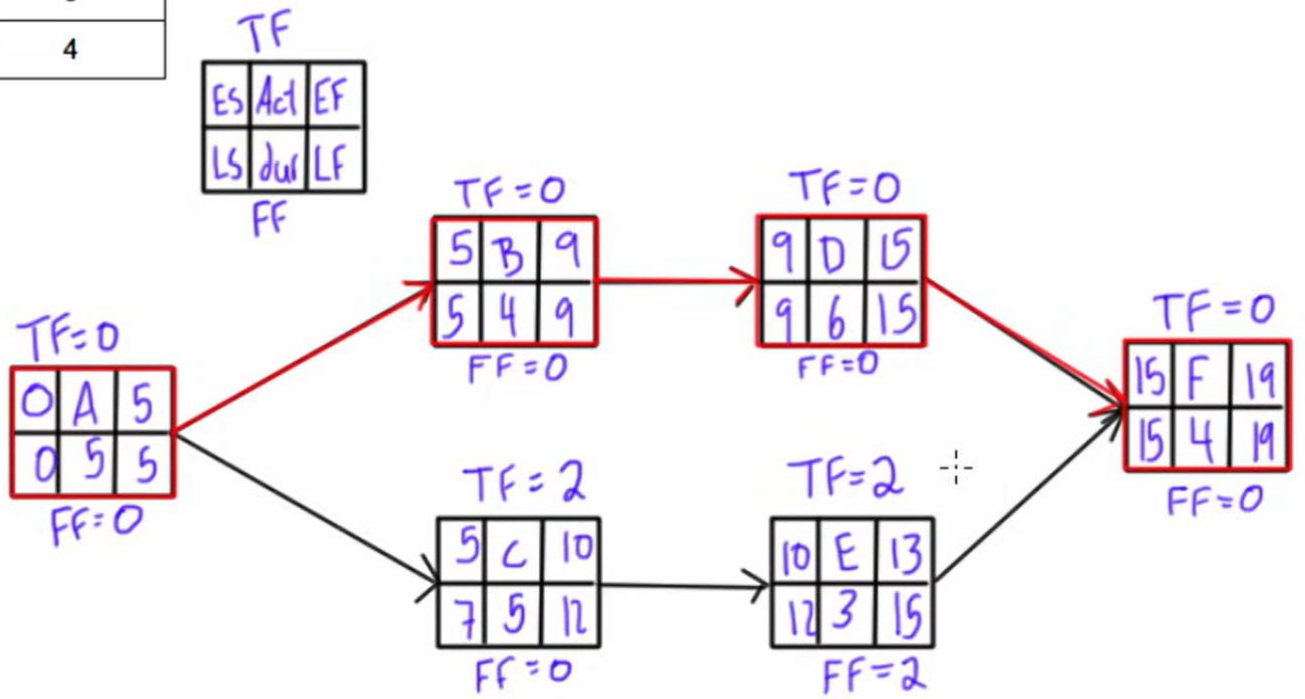
How to calculate total slack https://www.youtube.com/watch?v=TQcJ4bJDhQQ

# What is the critical path and project duration?
# What are ES, EF, LS, LF, TS, FS of these activities?

| Activity | Predecessor | Duration |
|---|---|---|
| A | - | 5 |
| B | A | 4 |
| C | A | 5 |
| D | B | 6 |
| E | C | 3 |
| F | D,E | 4 |

$$\text{Free Float} = FF = \text{Minimum } ES_{successors} - ES_{Activity} - \text{Duration}_{Activity}$$

$$FF \leq TF$$




What is free slack (free float)
https://www.youtube.com/watch?v=ErsolVw9UKE&t=0s&index=5&list=PLP_0l6U5zhU6FZojgg_0G5SeRAVY9Ocvu

# Using the Critical Path to Shorten a Project Schedule

- Seeing the schedule, project sponsors may ask if it can take less time
- Schedule compression or shortening durations of critical activities/tasks by adding more resources or changing their scope
- Main techniques for shortening schedules
  - Crashing activities by obtaining the greatest amount of schedule compression for the least incremental cost
    - E.g ask staff to work full time, instead of part time, or hire a temporary staff
  - Fast tracking activities by doing them in parallel or overlapping them

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# Importance of Updating Critical Path Data

- It is important that the project manager updates the schedule with actual data
  - Note actual activity durations as they are completed
  - Revise estimates for activities in progress
  - Monitor changes to make informed decisions

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# Critical Chain Scheduling (1 of 7)

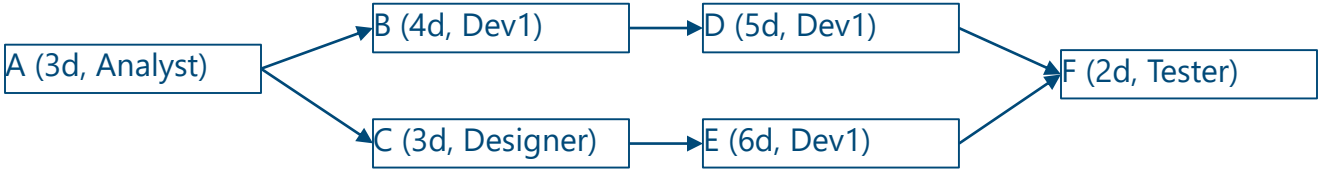
- Uses the Theory of Constraints (TOC): management philosophy developed by Eliyahu M. Goldratt
  - A chain and its weakest link metaphor – a chain is only as strong as its weakest link
  - Any complex system at any point in time often has only one aspect or constraint (bottleneck) that limits the ability to achieve more of the system's goal
- Similar to the Critical Path Method (CPM), but it explicitly considers resource constraints
- Critical Chain = longest path in the network diagram after considering both task dependencies and resource constraints
  - Whereas Critical Path = longest path in the network diagram based mainly on task dependencies
  - Resources used in the critical chain are critical resources

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# Critical Chain Scheduling (2 of 7)

| Task | Activity | Predecessor | Duration | Resource |
|---|---|---|---|---|
| A | Requirements | — | 3 days | Analyst |
| B | Database Design | A | 4 days | Developer 1 |
| C | UI Design | A | 3 days | Designer |
| D | Backend Development | B | 5 days | **Developer 1** |
| E | Frontend Development | C | 6 days | **Developer 1** |
| F | Integration & Test | D, E | 2 days | Tester |

## Critical Path Method




```
A → B → D → F = 3 + 4 + 5 + 2 = 14 days
A → C → E → F = 3 + 3 + 6 + 2 = 14 days
```

So both paths are **critical paths**, and the CPM project duration is **14 days**.

B, D, and E all require Developer1.

D and E would need Developer1 at overlapping times.

Developer1 cannot perform both simultaneously.

We must resource-level the schedule.

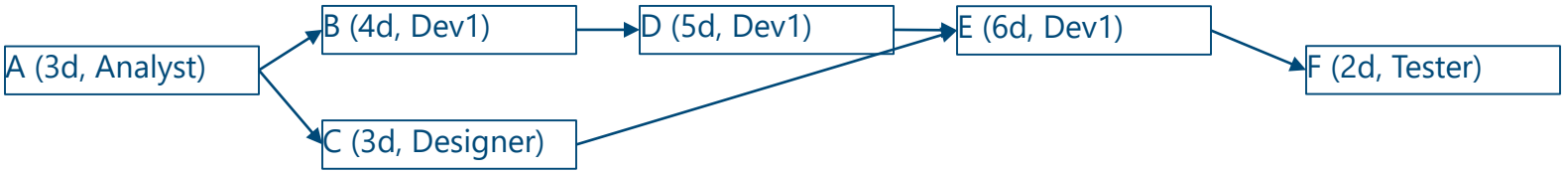
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# Critical Chain Scheduling (3 of 7)

## Critical Chain

Suppose we choose to perform the Developer1 activities in this order: B → D → E

The resource constraint effectively creates an additional relationship between D and E.




The resource-constrained sequence determining project completion is approximately:

**A → B → D → E → F**

with duration:

**3 + 4 + 5 + 6 + 2 = 20 days**

This is the **critical chain**.

The important point is that **D → E was not originally a technological/logical dependency**. We introduce their sequence because **both tasks require the same scarce resource, Developer 1**.

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# Critical Chain Scheduling (4 of 7)

- Critical Chain attempts to minimize multitasking when a resource works on more than one task at a time
  - Working on multiple tasks simultaneously introducing major delays in completing tasks

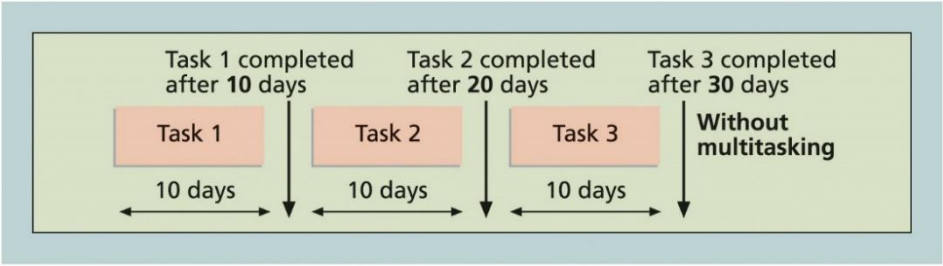



FIGURE 6-10a Three tasks without multitasking

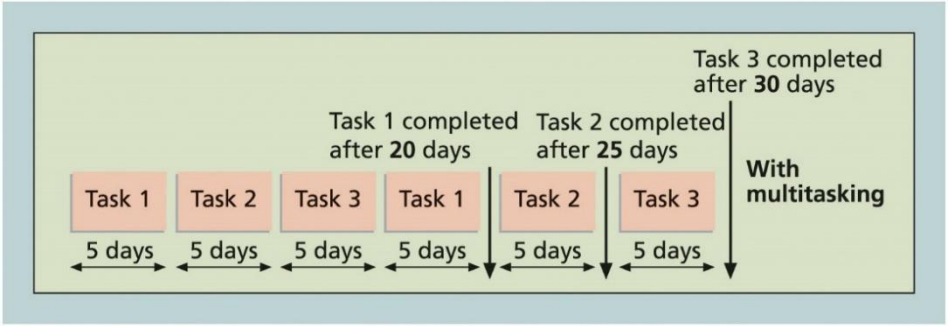



FIGURE 6-10b Three tasks with multitasking

  - Project is treated much like a relay race, with each task being completed as quickly as possible and then passed on to the next person

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# Critical Chain Scheduling (5 of 7)

- Uncertainty is not handled by giving every task a large safety margin for:
  - Negative effects of multitasking, distractions, interruptions
  - Murphy's law: If something can go wrong, it will
  - Parkinson's Law: Work expands to fill the time allowed
  - Student syndrome: Start work at the last possible moment before deadline
- Instead, task estimates are made more aggressive, and some of the removed safety is pooled into buffers.
  - For example, remove time (50%) from individual activities

https://pmstudycircle.com/critical-chain-method-ccm-in-project-management/



Critical chain vs critical path https://www.youtube.com/watch?v=mpc_FdAt75A

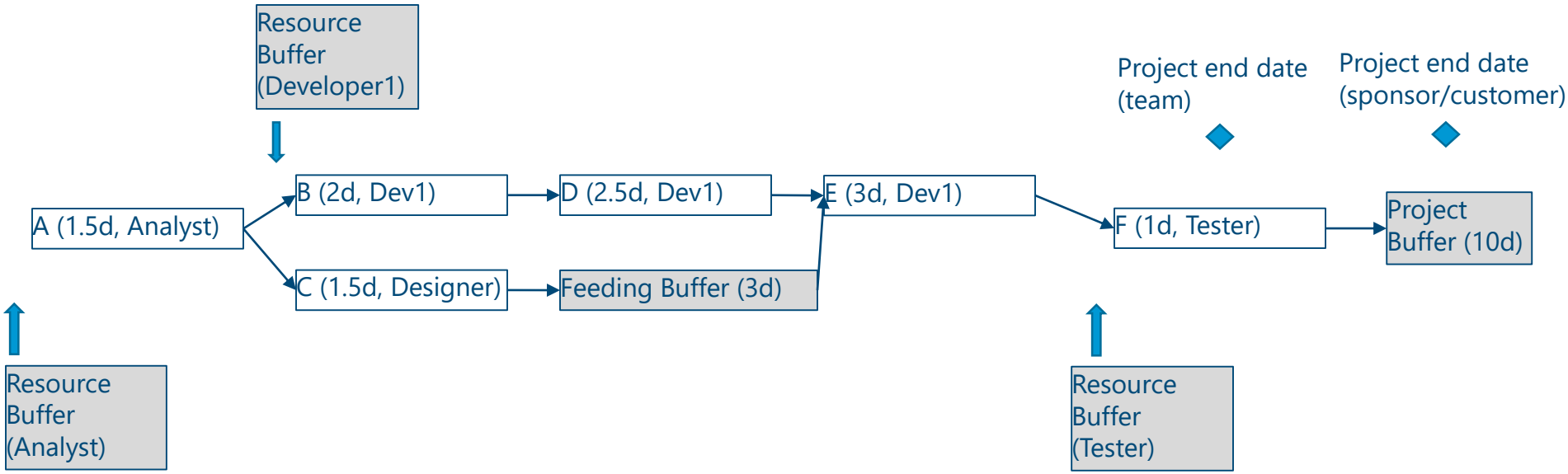
# Critical Chain Scheduling (6 of 7)

- Project buffer – Additional time placed at the end of the critical chain, but before the project end date.
  - Any delay on the critical chain will eat this buffer and the project end date will not change.
- Feeding buffers – Additional time inserted between the last task on a non-critical chain and the critical chain so that the duration is equal to the critical chain.
  - Any delay on the non-critical chain will eat this buffer and the critical chain will not be affected.
- Resource buffers – Readiness mechanisms (not time) kept alongside the critical chain to make sure that the critical resources are available to work on the critical chain activities.
  - Can be human resources or equipment.
  - For example, days before B is scheduled to begin, the resource buffer alerts the organization: "Developer1 will be needed for critical-chain Activity B in 3 days." This helps ensure that when A finishes, B can begin immediately rather than waiting for Developer1.
- If any activity finishes early, the time gain will be added to the buffer.

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# Critical Chain Scheduling (7 of 7)

Critical Chain




Project monitoring monitors buffer consumption. For example,
if the project is 50% completed

- 50% of project buffer remains -> OK
- 25% of project buffer remains -> problems occur and corrective actions are needed

# 6. Controlling the Schedule

- Goals of schedule control
  - Know the status of the schedule
  - Influence the factors that cause schedule changes
  - Determine that the schedule has changed
  - Manage changes when they occur
- Tools and techniques include
  - Earned value analysis (see chapter 7 project cost management)
  - Iteration burndown charts
  - Performance review
  - Critical path method
  - Project management information systems
  - Resource optimization such as resource leveling (see chapter 9 project resource management)
  - Schedule compression, e.g. crashing, fast tracking
  - Etc.

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# Scheduling and the Need for Discipline

- Important activities
  - Review the draft schedule or estimated completion date in the project charter
  - Prepare a more detailed schedule with the project team and get stakeholders' approval
  - Make sure the schedule is realistic and followed
  - Alert top management well in advance if there are schedule problems

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# Using Software to Assist in Project Schedule Management

- Software for facilitating communications helps people exchange schedule-related information
- Decision support models help analyze trade-offs that can be made to address schedule issues
- Project management software can help in various time management areas

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# Considerations for Agile/Adaptive Environments

- Schedule management is radically different using Agile and Scrum
  - Projects that rely heavily on the critical path method consider meeting the project's estimated completion date as a crucial component of success
  - Agile projects may not even need to estimate activity durations or project schedules; less focus on activity durations
    - The team can choose appropriate amount of work to fit each iteration
    - Product owners/customers are empowered to demand what they want in each iteration and they get to see, touch, and approve a working product.

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# Chapter Summary

- Project time management is often cited as the main source of conflict on projects
  - Most IT projects exceed time estimates
- Main processes
  - Plan schedule management
  - Define activities
  - Sequence activities
  - Estimate activity durations
  - Develop schedule
  - Control schedule

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